

# On the Schrödinger Equation

Kieron Eisma

August 2026

## Abstract

The Schrödinger Equation is an indispensable postulate of quantum mechanics that governs the evolution of the wave function:

$$i\hbar \frac{\partial \psi(\vec{r}, t)}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi(\vec{r}, t) + \hat{V}(\vec{r}, t) \psi(\vec{r}, t).$$

Because it is taken as a postulate in quantum theory, one cannot derive the Schrödinger Equation in the "same manner" as other equations in Physics, such as, for instance, the Euler-Lagrange Equation.

This begs the question of *why* the Schrödinger Equation was chosen as a postulate, instead of some other relation, and this is what the paper aims to answer.

Errors found in this paper, queries, along with suggestions are most welcome.

kiwi.eisma@gmail.com

## Contents

|          |                                 |          |
|----------|---------------------------------|----------|
| <b>1</b> | <b>The Wave Packet</b>          | <b>1</b> |
| <b>2</b> | <b>The Schrödinger Equation</b> | <b>2</b> |
|          | <b>References</b>               | <b>3</b> |

## 1 The Wave Packet

*All wave functions in this paper are assumed to be of moderate decrease on the real line, and operators have not been explicitly indicated.*

The wave equation is

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u.$$

We shall consider the one-dimensional case for simplicity; the solutions to this equation yield plane waves that propagate along the  $x$ -axis. The general form of a wave packet is the superposition of these solutions:

$$\psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(k) e^{i(kx - \omega t)} dk. \quad (1)$$

Define the initial wave function  $\psi_0(x) = \psi(x, 0)$ , so that

$$\psi_0(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(k) e^{ikx} dx,$$

where, by the Fourier Inversion Formula,

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi_0(x) e^{-ikx} dx.$$

The de Broglie Relation implies the momentum is given by  $p = \hbar k$ , where  $k$  is the wave vector ( $k = \vec{k}$  in  $\mathbb{R}^1$ ). It follows that  $dk = \frac{dp}{\hbar}$ . Moreover, the Planck Relation yields the energy:  $E = \hbar\omega$ . Therefore, Equation 1 becomes

$$\psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi\left(\frac{p}{\hbar}\right) e^{i\left(\frac{p}{\hbar}x - \frac{E}{\hbar}t\right)} \frac{dp}{\hbar}.$$

Now, define  $\tilde{\phi}(p) = \frac{1}{\sqrt{\hbar}} \phi\left(\frac{p}{\hbar}\right) = \frac{1}{\sqrt{\hbar}} \phi\left(\frac{p}{\hbar}\right)$ . Then, we may express the wave packet above in the form

$$\begin{aligned} \psi(x, t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \sqrt{\hbar} \tilde{\phi}(p) e^{i\left(\frac{p}{\hbar}x - \frac{E}{\hbar}t\right)} \frac{dp}{\hbar} \\ &= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \tilde{\phi}(p) e^{\frac{i}{\hbar}(px - Et)} dp. \end{aligned}$$

## 2 The Schrödinger Equation

In Section 1, we showed that the equation of a wave packet in  $\mathbb{R}^1$  is given by the equation<sup>1</sup>

---

<sup>1</sup>Recall that the kinetic energy is given by  $T = \frac{p^2}{2m}$ .

$$\psi(x, t) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \tilde{\phi}(p) \exp \left[ \frac{i}{\hbar} \left( px - \left( \frac{p^2}{2m} + V \right) t \right) \right] dp;$$

It follows that

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = \frac{1}{\sqrt{2\pi\hbar}} \left( i\hbar \frac{\partial}{\partial t} \int_{-\infty}^{\infty} \tilde{\phi}(p) \exp \left[ \frac{i}{\hbar} \left( px - \left( \frac{p^2}{2m} + V \right) t \right) \right] dp \right) \quad (2)$$

$$= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \tilde{\phi}(p) \left( \frac{p^2}{2m} + V \right) \exp \left[ \frac{i}{\hbar} \left( px - \left( \frac{p^2}{2m} + V \right) t \right) \right] dp \quad (3)$$

$$= \left( \frac{p^2}{2m} + V \right) \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \tilde{\phi}(p) \exp \left[ \frac{i}{\hbar} \left( px - \left( \frac{p^2}{2m} + V \right) t \right) \right] dp \quad (4)$$

$$= \left( \frac{p^2}{2m} + V \right) \psi(x, t). \quad (5)$$

However, recall that the momentum operator in position space is given by<sup>2</sup>

$$\vec{p} = -i\hbar \vec{\nabla},$$

which is to say  $p^2 = \hbar^2 \nabla^2$ , so in  $\mathbb{R}^1$ ,

$$p^2 = \hbar^2 \frac{\partial^2}{\partial x^2}.$$

Substituting this into Equation 5,

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} \psi(x, t) &= \left( \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V \right) \psi(x, t) \\ &= \hat{H} \psi(x, t), \end{aligned}$$

where  $\hat{H}$  denotes the Hamiltonian. This is the (time-dependent) one-dimensional Schrödinger Equation.

## References

[Zet22] Nouredine Zettili. *Quantum Mechanics*. Wiley, 2022. ISBN: 978-1-118-30789-2.

---

<sup>2</sup>See Equation 2.346 in [Zet22]